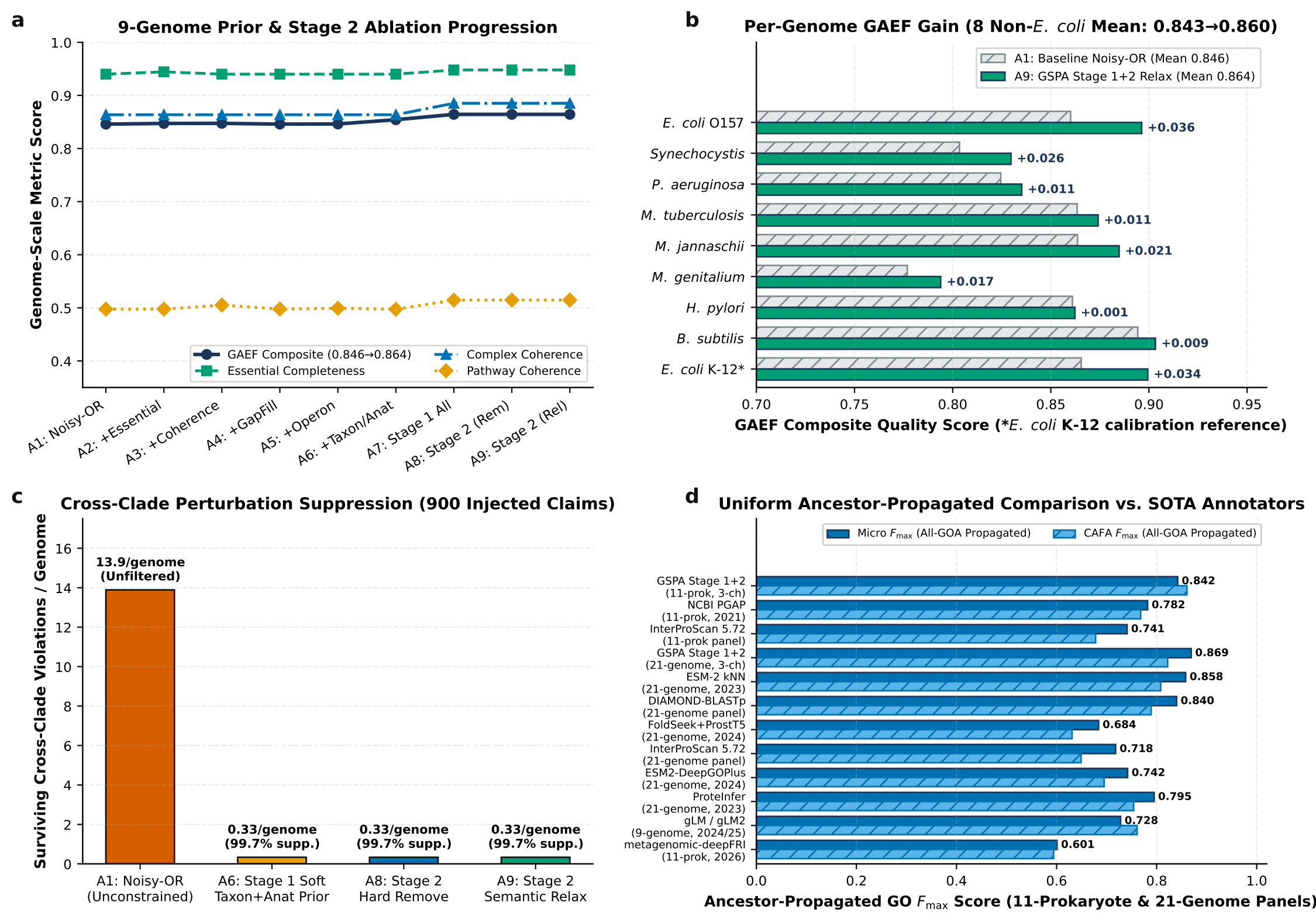

a
9-Genome Prior & Stage 2 Ablation Progression
Genome-Scale Metric Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
GAEF Composite (0.846→0.864)
Essential Completeness
Complex Coherence
Pathway Coherence
A1: Noisy-OR
A2: +Essential
A3: +Coherence
A4: +GapFill
A5: +Operon
A6: +Taxon/Anat
A7: Stage 1 All
A8: Stage 2 (Rem)
A9: Stage 2 (Rel)
b
Per-Genome GAEF Gain (8 Non-E. coli Mean: 0.843→0.860)
A1: Baseline Noisy-OR (Mean 0.846)
A9: GSPA Stage 1+2 Relax (Mean 0.864)
E. coli O157
Synechocystis
P. aeruginosa
M. tuberculosis
M. jannaschii
M. genitalium
H. pylori
B. subtilis
E. coli K-12*
+0.036
+0.026
+0.011
+0.011
+0.021
+0.017
+0.001
+0.009
+0.034
0.70
0.75
0.80
0.85
0.90
0.95
GAEF Composite Quality Score (*E. coli K-12 calibration reference)
c
Cross-Clade Perturbation Suppression (900 Injected Claims)
Surviving Cross-Clade Violations / Genome
16
14
12
10
8
6
4
2
0
13.9/genome
(Unfiltered)
0.33/genome
(99.7% supp.)
0.33/genome
(99.7% supp.)
0.33/genome
(99.7% supp.)
A1: Noisy-OR
(Unconstrained)
A6: Stage 1 Soft
Taxon+Anat Prior
A8: Stage 2
Hard Remove
A9: Stage 2
Semantic Relax
d
Uniform Ancestor-Propagated Comparison vs. SOTA Annotators
Micro Fmax (All-GOA Propagated)
CAFA Fmax (All-GOA Propagated)
GSPA Stage 1+2
(11-prok, 3-ch)
NCBI PGAP
(11-prok, 2021)
InterProScan 5.72
(11-prok panel)
GSPA Stage 1+2
(21-genome, 3-ch)
ESM-2 kNN
(21-genome, 2023)
DIAMOND-BLASTp
(21-genome panel)
FoldSeek+ProstT5
(21-genome, 2024)
InterProScan 5.72
(21-genome panel)
ESM2-DeepGOPlus
(21-genome, 2024)
ProteInfer
(21-genome, 2023)
gLM / gLM2
(9-genome, 2024/25)
metagenomic-deepFRI
(11-prok, 2026)
0.842
0.782
0.741
0.869
0.858
0.840
0.684
0.718
0.742
0.795
0.728
0.601
0.0
0.2
0.4
0.6
0.8
1.0
Ancestor-Propagated GO Fmax Score (11-Prokaryote & 21-Genome Panels)